

Math 343 / 643 Spring 2026

Midterm Examination Two

Professor Adam Kapelner

April 22, 2026

Full Name _____

Code of Academic Integrity

Since the college is an academic community, its fundamental purpose is the pursuit of knowledge. Essential to the success of this educational mission is a commitment to the principles of academic integrity. Every member of the college community is responsible for upholding the highest standards of honesty at all times. Students, as members of the community, are also responsible for adhering to the principles and spirit of the following Code of Academic Integrity.

Activities that have the effect or intention of interfering with education, pursuit of knowledge, or fair evaluation of a student's performance are prohibited. Examples of such activities include but are not limited to the following definitions:

Cheating Using or attempting to use unauthorized assistance, material, or study aids in examinations or other academic work or preventing, or attempting to prevent, another from using authorized assistance, material, or study aids. Example: using an unauthorized cheat sheet in a quiz or exam, altering a graded exam and resubmitting it for a better grade, etc.

I acknowledge and agree to uphold this Code of Academic Integrity.

signature

date

Instructions

This exam is 75 minutes (variable time per question) and closed-book. You are allowed **one** page (front and back) of a “cheat sheet”, blank scrap paper (provided by the proctor) and a graphing calculator (which is not your smartphone). Please read the questions carefully. Within each problem, I recommend considering the questions that are easy first and then circling back to evaluate the harder ones. No food is allowed, only drinks.

Problem 1 Consider the following full-rank design matrix:

$$\mathbf{X} := [\mathbf{1}_n \mid \mathbf{x}_{\cdot 1} \mid \dots \mid \mathbf{x}_{\cdot p}] = \begin{bmatrix} \mathbf{x}_{1\cdot} \\ \vdots \\ \mathbf{x}_{n\cdot} \end{bmatrix}$$

with column indices $0, 1, \dots, p$ and row indices $1, 2, \dots, n$. And let $\mathbf{H}$ be the orthogonal projection matrix (which is idempotent and symmetric) onto the column space of $\mathbf{X}$. We assume also a continuous (real-valued) response model which is linear in these measurements, i.e. $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\mathcal{E}}$. For the error term, we assume the “core assumption”,

$$\boldsymbol{\mathcal{E}} \sim \mathcal{N}_n(\mathbf{0}_n, \sigma^2 \mathbf{I}_n) \quad \text{where } \sigma^2 > 0 \text{ and thus } \frac{1}{\sigma^2} \boldsymbol{\mathcal{E}}^\top \boldsymbol{\mathcal{E}} \sim \chi_n^2.$$

Consider the following estimator for $\boldsymbol{\beta}$: $\mathbf{B} := (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and let $\hat{\mathbf{Y}} := \mathbf{X}\mathbf{B}$ and $\mathbf{E} := \mathbf{Y} - \hat{\mathbf{Y}}$. We established in class that $\mathbf{B} \sim \mathcal{N}_{p+1}(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$. We also established that $\mathbf{B}$ and $\mathbf{E}$ are independent. Let $\boldsymbol{\epsilon}$ denote the realization from $\boldsymbol{\mathcal{E}}$.

- (a) [2 pt / 2 pts] Is $\mathbf{B}$ an unbiased estimator for $\boldsymbol{\beta}$? Circle one: Yes / No
- (b) [2 pt / 4 pts] Is $\mathbf{B}$ the maximum likelihood estimator for $\boldsymbol{\beta}$? Circle one: Yes / No
- (c) [2 pt / 6 pts] Is $\mathbf{E}$ an unbiased estimator estimator for σ^2 ? Circle one: Yes / No
- (d) [2 pt / 8 pts] Does $\mathbb{E}[\boldsymbol{\mathcal{E}} - \mathbf{E}] = \mathbf{0}_n$? Circle one: Yes / No
- (e) [5 pt / 13 pts] Write the z test statistic for testing $H_0 : \beta_{17} = 37$ assuming σ^2 is known.

- (f) [5 pt / 18 pts] Find the distribution of $\hat{\mathbf{Y}}$ from the information given in the problem header and results from MATH 340. Simplify as much as possible.

- (g) [8 pt / 26 pts] Prove that $\boldsymbol{\mathcal{E}}^\top(\mathbf{I}_n - \mathbf{H})\boldsymbol{\mathcal{E}} = \text{SSE}$. Justify each step.

Problem 2 Consider the Boston Housing Data which has $n = 506$ and response `medv` with $\bar{y} = 22.53$ and $s_y = 9.20$. We assume the linear model is true and model `medv` using OLS on `zn + rm + nox + dis + lstat`, all continuous (non-categorical) features. Below is the $(\mathbf{X}^T \mathbf{X})^{-1}$ where $\mathbf{X}$ is the design matrix:

	(Intercept)	zn	rm	nox	dis	lstat
(Intercept)	0.58000	4.4e-04	-5.1e-02	-2.7e-01	-1.8e-02	-3.0e-03
zn	0.00044	6.9e-06	-4.5e-05	-7.1e-05	-5.1e-05	-2.0e-07
rm	-0.05100	-4.5e-05	6.9e-03	7.5e-04	6.3e-04	4.4e-04
nox	-0.27000	-7.1e-05	7.5e-04	4.2e-01	1.5e-02	-1.9e-03
dis	-0.01800	-5.1e-05	6.3e-04	1.5e-02	1.5e-03	4.3e-05
lstat	-0.00300	-2.0e-07	4.4e-04	-1.9e-03	4.3e-05	8.9e-05

The RMSE for this regression is 5.289 and here are the slope estimates:

(Intercept)	zn	rm	nox	dis	lstat
16.14	0.06	4.44	-15.20	-1.44	-0.66

- (a) [5 pt / 31 pts] Assume that errors are independent. Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable `nox`? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.
- (b) [5 pt / 36 pts] Assume that errors are independent, mean zero and the errors are homoskedastic. Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable `nox`? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

- (c) [5 pt / 41 pts] Assume that errors are independent, mean zero and the errors are heteroskedastic. Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable **nox**? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

- (d) [5 pt / 46 pts] Assume that errors are *normal*, independent, mean zero and the errors are heteroskedastic. Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable **nox**? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

- (e) [5 pt / 51 pts] Assume the core assumption. But now consider that the estimates in the problem header came from a ridge regression with $\lambda = 1$. Given the information above and the facts from problem 1, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable `nox`? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

- (f) [5 pt / 56 pts] Assume the $\mathbf{X}$ matrix's columns were standardized so that the average of each column was zero and the standard deviation of each column was 1. Let $\lambda = 1000$. Would the slope coefficient estimates found in the problem header be expected for a lasso regression fit? Why or why not?

- (g) [5 pt / 61 pts] Assume the core assumption. Compute $\mathbb{P}(R^2 > 0.9)$ under H_0 that all features in the model `zn`, `r`, `nox`, `dis`, `lstat` do not have any linear effect. You will need to express your answer using `pbeta( $x, \alpha, \beta$ )`, the CDF of the beta distribution. But each argument (i.e. x, α, β) must be numeric.

Problem 3 Consider the following survival distribution:

$$Y \sim \text{Lomax}(\alpha, \beta) := \frac{\alpha}{\beta} \left(1 + \frac{y}{\beta}\right)^{-(\alpha+1)} \mathbf{1}_{y>0}$$

We will assume a restricted parameter space $\alpha > 2$, $\beta > 0$ so that the expectation and variance always exists:

$$\mathbb{E}[Y] = \frac{\beta}{\alpha - 1}, \quad \mathbb{V}\text{ar}[Y] = \frac{\beta^2 \alpha}{(\alpha - 1)^2 (\alpha - 2)}$$

We would like to reparameterize the Lomax so that $\mathbb{E}[Y] = \theta > 0$. We do so below:

$$Y \sim \text{Lomax}(\alpha, \theta) := \frac{\alpha}{\theta(\alpha - 1)} \left(1 + \frac{y}{\theta(\alpha - 1)}\right)^{-(\alpha+1)} \mathbf{1}_{y>0}$$

We now wish to use a GLM to estimate p slope coefficients $\boldsymbol{\beta}$ using data from a matrix $\mathbf{X}$ defined in the problem 1 header.

- (a) [5 pt / 66 pts] What is an example link function ϕ we can use to connect θ to $\mathbf{x}_i \boldsymbol{\beta}$ and why would your example work in this context?

Regardless of your answer to the previous problem, employ $\theta = \phi(\mathbf{x}_i, \boldsymbol{\beta}) = e^{\mathbf{x}_i \cdot \boldsymbol{\beta}}$ going forward. Also assume $Y_1, \dots, Y_n \stackrel{iid}{\sim} \text{Lomax}(\alpha, \theta)$ as defined in the problem header.

- (b) [7 pt / 73 pts] Write an expression below that would compute the $\mathbf{B}$, maximum likelihood estimator (not estimate) for vector $\boldsymbol{\beta}$ and A , the maximum likelihood estimator (not estimate) for α as a function of $\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{Y}$. You do not need to solve. You do not need to simplify. Assume there is *no censoring*. There are many equivalent expressions that answer this question correctly.

- (c) [3 pt / 76 pts] Are there any nuisance parameter(s) in this estimation? If yes, list the nuisance parameter(s). If no, write “none”.

We employ the correct MLE for $\mathbf{B}, A$ going forward.

- (d) [2 pt / 78 pts] If there was censoring in $Y_1, \dots, Y_n$ would the MLE expression in (b) change? Circle one: Yes / No
- (e) [2 pt / 80 pts] Is A asymptotically normal? Circle one: Yes / No
- (f) [2 pt / 82 pts] Is $\mathbb{V}\text{ar}[A] = \mathbb{V}\text{ar}[Y]$? Circle one: Yes / No

We now fit the lung survival data on the uncensored observations. The response variable is days of survival. We find the following results:

	Value	Std. Error	z	p
(Intercept)	7.17e+00	1.08e+00	6.64	3.2e-11
inst	2.05e-02	8.80e-03	2.32	0.0201
age	-7.54e-03	8.06e-03	-0.94	0.3497
sex	3.91e-01	1.39e-01	2.82	0.0048
ph.ecog	-6.26e-01	1.58e-01	-3.95	7.7e-05
ph.karno	-1.86e-02	7.67e-03	-2.43	0.0151
pat.karno	7.68e-03	5.54e-03	1.39	0.1656
meal.cal	7.32e-06	1.81e-04	0.04	0.9678
wt.loss	1.10e-02	5.34e-03	2.07	0.0387
Loglik(model)= -824.2 Loglik(intercept only)= -841.1				

- (g) [6 pt / 88 pts] Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{age}}, 95\%}$, the confidence interval for the true slope parameter of the variable **age**? If yes, then compute it below to three significant digits and indicate if the CI is approximate or exact. If no, explain why.

- (h) [7 pt / 95 pts] Test the omnibus null at the 5% significance level. Justify each step. Here are some values of the inverse CDF of the χ^2_{df} distribution:

df	Probability less than the critical value				
	0.90	0.95	0.975	0.99	0.999
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458
7	12.017	14.067	16.013	18.475	24.322
8	13.362	15.507	17.535	20.090	26.124

- (i) [5 pt / 100 pts] Predict the number of days of survival for someone with $\text{inst} = 1$, $\text{age} = 75$ but all other measurements are zero.